

Broadband Vernier-Programmable Thin-Film Lithium Niobate Triple-Ring Electro-Optic Modulator for High-Speed Optical Interconnects

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Abstract We demonstrate a broadband Vernier-programmable triple-ring lithium niobate electro-optic modulator that overcomes single-ring high-frequency efficiency roll-off through supermode coupling. DC bias enables 19.4-GHz tuning of the peak RF operating frequency. The extracted response indicates a sideband-to-carrier ratio from -1.4 to -2.7 dB over this range. ©2026 The Author(s)

Introduction

The explosive growth of large language models (LLMs) has driven computing demand far beyond what chip integration alone can sustain [1-3], making optical interconnects a more promising solution for large-scale data transfer and higher system throughput. The data rate of optical interconnects critically depends on the bandwidth of electro-optic (EO) modulators in PICs. Compared with Mach-Zehnder modulators, micro-ring modulators offer a compact footprint and avoid the velocity-matching requirement of traveling-wave electrodes [4]. However, their resonant nature leads to a trade-off between EO conversion efficiency and modulation frequency, limiting the modulation bandwidth. To overcome this limitation, previous works have employed coupled double rings and their symmetric and antisymmetric supermodes to enable energy transfer between the laser and the optical sidebands [5-10]. However, the resonance splitting is determined by the inter-ring coupling strength which is fixed by the device geometry and layout. Such devices can only achieve high efficiency near a specific

frequency rather than over a broad frequency range. Tunable approaches such as MEMS actuation have been explored [11-13], but they usually come at the cost of increased fabrication complexity and reduced device reliability.

In this work, we design and characterize a broadband Vernier-programmable triple-ring EO modulator. The fabricated device exhibits three well-resolved resonances, whose measured DC-bias evolution is consistent with the expected supermode behavior. By inserting a third Vernier ring between two outer rings, the proposed device introduces an additional degree of freedom for tuning the resonance splitting. As a result, the peak-efficiency RF operating frequency can be programmed from 51.9 to 71.3 GHz, corresponding to a 19.4-GHz high-efficiency operating window. The extracted device response indicates a sideband-to-carrier ratio from -1.4 to -2.7 dB over this range, suggesting an EO bandwidth of 76.1 GHz. We further illustrate a triple-ring-based optical-interconnect architecture that could support an aggregate throughput of 18.25 Tb/s.

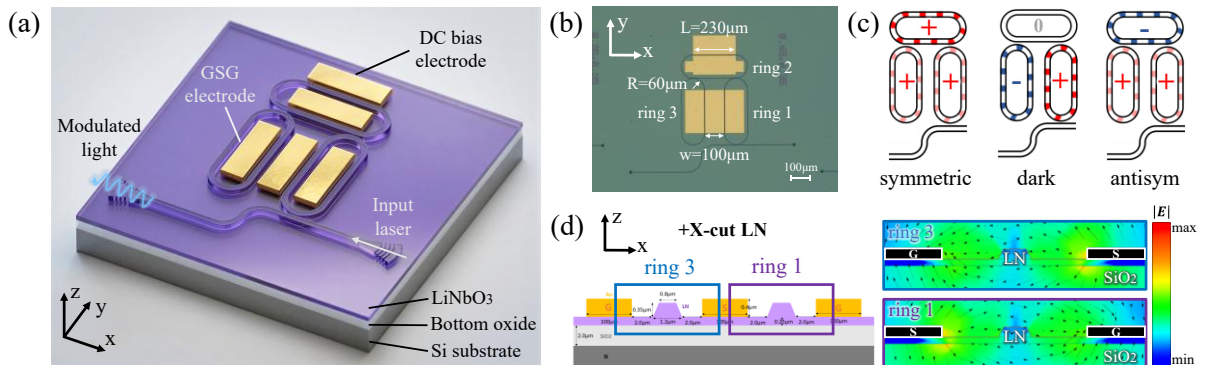


Fig. 1: (a) Schematic of the proposed triple-ring EO modulator. (b) Optical microscope image of the fabricated device. (c) Supermodes of the coupled triple-ring structure: symmetric, dark, and antisymmetric mode. Red/blue indicate opposite optical phases; gray indicates near-zero amplitude. (d) Cross-sectional electric-field distributions around the GSG electrodes for two outer rings.

Device Design

Fig. 1(a) illustrates the schematic of the Vernier-programmable triple-ring modulator demonstrated in this work, and Fig. 1(b) shows an optical micrograph of the fabricated device. The device is fabricated on an x-cut LNOI platform. All three resonators adopt a racetrack geometry and they are evanescently coupled through gaps, while TE mode light is coupled in and out through grating couplers at the two ends of the bus waveguide. Two sets of gold electrodes are integrated outside the resonators, including GSG electrodes and DC bias electrodes. Through the Pockels effect in LN, the applied electric field modulates the refractive index and thus the optical resonances. Specifically, the GSG electrodes apply RF signals to the straight sections of the two outer rings and provide opposite-phase push-pull modulation through the r_{33} coefficient, while the DC electrodes tune the straight section of the Vernier ring through the r_{13} coefficient. As shown in Fig. 1(d), this push-pull configuration enhances the modulation efficiency of the two outer racetracks without requiring two RF signals with opposite phases [14-16].

According to coupled-oscillator theory, the triple-ring structure supports three supermodes: symmetric, dark, and antisymmetric modes [10], as shown in Fig. 1(c). Opposite-phase modulation of the two outer rings enables energy transfer between the dark mode and the symmetric/antisymmetric modes. When the resonance of the middle Vernier ring is tuned, the dark mode remains unchanged, while the symmetric and antisymmetric modes shift at different rates, as shown in Fig. 2(c). Therefore, when the input

laser is aligned with the dark mode and the RF frequency matches the splitting frequency, the generated optical sidebands overlap with the adjacent symmetric and antisymmetric modes, enabling efficient energy transfer from the laser to the sidebands. As a result, electrical tuning of the Vernier ring enables programmable control of the supermode splitting, allowing high EO modulation efficiency to be sustained over a broad frequency range.

Fabrication

A lithium-niobate-on-insulator (LNOI) wafer with a 600 nm LN layer and a 2 μm buried oxide was used for device fabrication. The modulator was implemented using partially etched rib waveguides with a 350 nm etch depth and a 250 nm slab. Electron-beam lithography defined race-track rings with a 60 μm radius, a 230 μm straight section, a 0.8 μm waveguide width, and a 0.45 μm bus-to-ring gap. The waveguides were formed by Ar inductively coupled plasma etching and wet polishing. The electrodes were then fabricated by electron-beam lithography, 400 nm Ti/Au deposition, and lift-off [17].

Characterization

Fig. 2(a)(b) shows the characterization setup. Light was coupled into and out through optical fibers and grating couplers, while the DC bias and RF signals were applied to the corresponding electrodes. To examine the effect of DC bias on the supermode resonance splitting, a tunable laser source (TLS) was operated in wavelength-sweeping mode as the input, and the output transmission was recorded by a power meter to

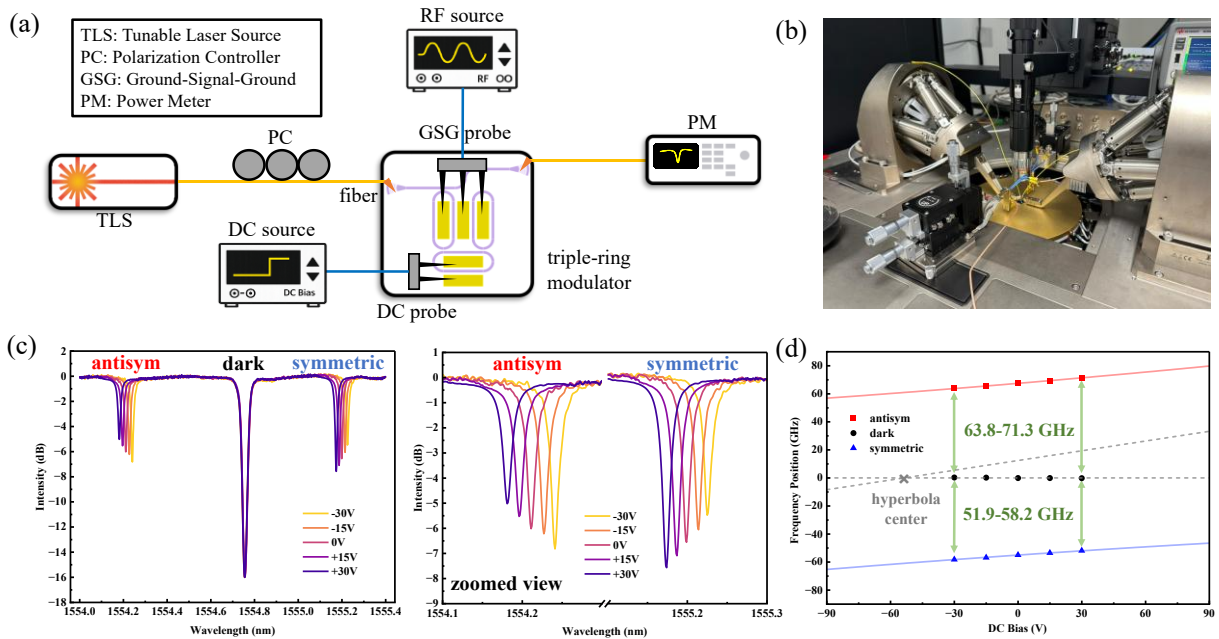


Fig. 2: (a) Schematic of the characterization setup for DC and EO measurements. (b) Photograph of the measurement setup. (c) Measured transmission spectra under different DC bias conditions, including zoomed views of the side resonances. (d) Relative frequency positions of the three resonances as functions of DC bias.

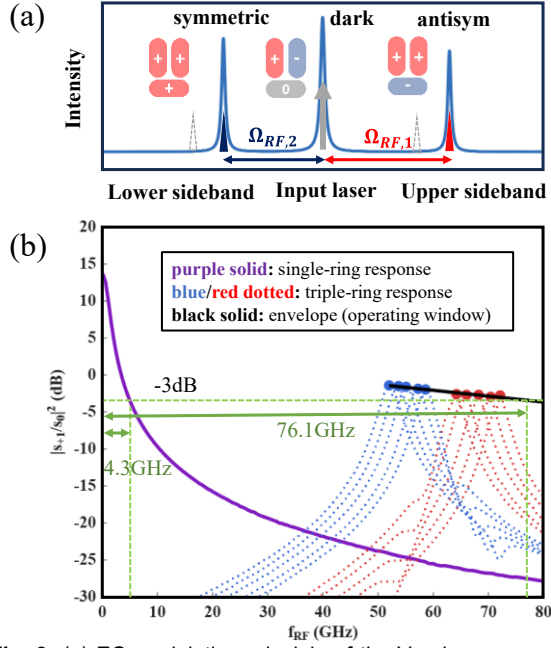


Fig. 3: (a) EO modulation principle of the Vernier-programmable triple-ring modulator under Vernier-ring detuning. (b) EO modulation efficiency of the triple-ring modulator obtained from the compact model.

obtain the transmission spectra under different DC bias conditions.

Fig. 2(c) shows the transmission spectra of the EO modulator from 1554.0 to 1555.4 nm under different DC bias voltages from -30 V to +30 V. The dark-mode resonance exhibits an extinction ratio of 16 dB with $Q_{\text{dark}} = 6.0 \times 10^4$, while the symmetric and antisymmetric resonances show lower extinction ratios of 5.4 - 7.7 dB with $Q_{\text{sym}} = 1.3 \times 10^5$ and $Q_{\text{anti}} = 1.2 \times 10^5$. These differences in Q factor and extinction ratio arise from the distinct participation of the three rings in each super-mode. Under DC bias tuning, the dark-mode resonance remains nearly unchanged, whereas the symmetric and antisymmetric resonances shift at different rates. As shown in Fig. 2(d), the resonance positions of these two modes follow a hyperbolic dependence on the applied DC bias, indicating that the supported modulation frequency of the proposed EO modulator can be programmably tuned from 51.9 to 71.3 GHz.

The EO modulation mechanism is illustrated in Fig. 3(a). The input laser is aligned to the center of the dark mode, and the modulation efficiency reaches its maximum when the RF frequency matches the frequency spacing between the modes. We define the EO modulation efficiency G as the ratio of the sideband power to the carrier power [18-19]. As shown in Fig. 3(b), with the RF drive amplitude fixed at 1.5 V, the device response extracted from the DC measurements indicates that the proposed modulator can maintain an EO modulation efficiency between -

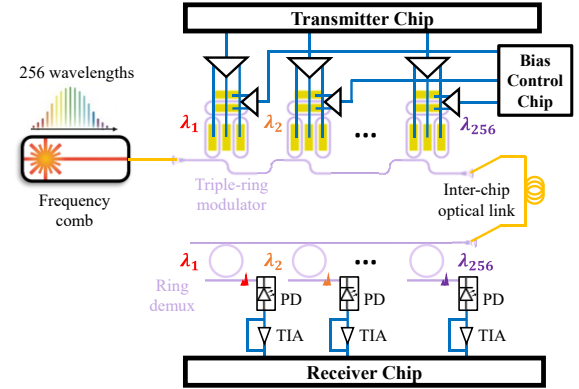


Fig. 4: Conceptual 256-wavelength WDM optical-interconnect architecture employing the proposed triple-ring modulator for parallel chip-to-chip communication.

1.4 and -2.7 dB over 51.9 to 71.3 GHz. Compared with a conventional single-ring modulator of the same structure (purple curve), the 3-dB bandwidth is increased from 4.3 to 76.1 GHz.

Interconnect Architecture

To highlight the system-level relevance of the proposed modulator, we further illustrate a WDM optical interconnect architecture for chip-to-chip communication, as shown in Fig. 4. Multiple laser carriers are multiplexed onto a common optical bus, modulated by parallel triple-ring modulators, and recovered by demultiplexing and photodetection at the receiver side. Assuming 256 wavelength channels operated in parallel with NRZ signaling, the architecture could in principle provide an aggregate throughput of 18.25 Tb/s.

Conclusions

We demonstrated a Vernier-programmable triple-ring EO modulator on x-cut LNOI. By electrically tuning the Vernier ring, programmable super-mode splitting is achieved, shifting the peak EO response from 51.9 to 71.3 GHz. The extracted response indicates an EO modulation efficiency from -1.4 to -2.7 dB over this 19.4-GHz range. Compared with a single-ring counterpart, the 3-dB bandwidth is increased from 4.3 to 76.1 GHz. These results highlight the proposed modulator as a compact candidate for high-speed photonic interconnects, with potential aggregate NRZ throughput up to 18.25 Tb/s in a 256-wavelength WDM architecture.

Acknowledgements

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